

# WASP-52b

R-0311

Benchmark run · workflow W8-benchmark-deep · record deep-bench\_WASP-52\_221220 · created 2026-07-17T23:07:24+00:00

BENCHMARK: ACCURATE — fit consistent with published values

## Results

|                                                   |                                 |
|---------------------------------------------------|---------------------------------|
| Mid-Transit Time (Tmid)                           | 2459933.6629 +/- 0.0018 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*)         | 0.176 +/- 0.011                 |
| Transit depth (Rp/Rs)^2                           | 3.09 +/- 0.38 [%]               |
| Orbital Inclination (inc)                         | 85.56 +/- 0.83                  |
| Airmass coefficient 1 (a1)                        | 1.279 +/- 0.014                 |
| Airmass coefficient 2 (a2)                        | -0.1148 +/- 0.0026              |
| Scatter in the residuals of the lightcurve fit is | 1.07 %                          |
| Best Comparison Star                              | None                            |
| Optimal Aperture                                  | 4.67                            |
| Optimal Annulus                                   | 15.29                           |
| Transit Duration (day)                            | 0.0777 +/- 0.0044               |

## Accuracy vs published ephemeris (ExoClock/NEA)

|                           |                                           |
|---------------------------|-------------------------------------------|
| Rp/R■ fit vs published    | 0.176 ± 0.011 vs 0.1646 (deviation 1.04σ) |
| Duration fit vs published | 0.0777 d vs 0.0758 d (deviation 0.42σ)    |
| Mid-time O–C              | -1.3 min                                  |
| Depth SNR                 | 16.0                                      |
| Residual scatter          | 1.07 %                                    |

## Light curve

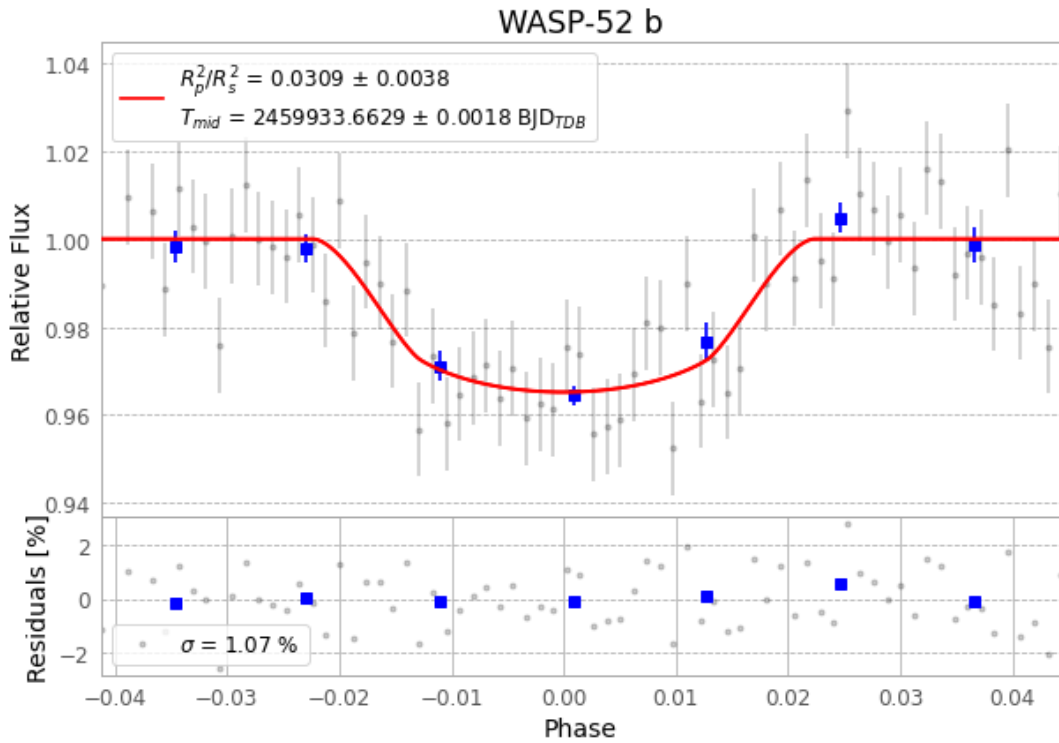


Figure 1 — FinalLightCurve\_WASP-52 b\_2022-12-19.png (SHA-256 65bb2f93368eef16...)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [315, 252], 3 comparison star(s). Exact configuration: parameters.inits\_json in record.json (SHA-256 efd09ae859d882a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline\_commit 1bbef5c

## Data provenance

74 FITS frames (49 MB), WASP-52221220020945.FITS → WASP-52221220055415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-221220.log (5b470300c358...), FinalLightCurve\_WASP-52 b\_2022-12-19.png (65bb2f93368e...), FinalLightCurve\_WASP-52 b\_2022-12-19.csv (de7988ecfe79...)

## Compute resources

Wall time 120.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 23:07Z.